

Table 2
Previous studies regarding the effect of surface roughness on CHF for metal and silicon surfaces.

Surface type	Author	Surface	Fluid	Roughness parameter	Roughness range	Wettability	θ	Effect on CHF	Correlation
Metal	Ramilison et al. [15] [*]	Copper	Various	Rms roughness, R_q R_a	2–12 μm 0.02–1.5 μm	Various	0–40°	±10–20% enhancement	$\frac{q''_{\text{CHF}}}{q''_{\text{CHF},Z}} = 0.0336(\pi - \theta_r)^{3.0}(R_q)^{0.125}$
	Ferjančič and Golobič [16,28]	SS 302, steel 1010	FC-72, water			Not reported	–	Max 51% increase (etched surface)	$q''_{\text{CHF}} = a \ln(R_a) + b$
	Ahmad et al. [17]	Copper	R-123	P_a (R_a without cut-off filter)	3.5 μm	Highly wetting fluid	–	225 kW/m ²	No correlation
	Guan et al. [18] ^{**}	Brass	Pentane, hexane, FC-72	R_q	0.15–5 μm	Highly wetting fluids	–	Increases 1.15 times	$\frac{q''_{\text{CHF}}}{q''_{\text{CHF},\text{smooth}}} = \sqrt{\frac{A^2 + \sqrt{A^4 + 4}}{2}}$
Si wafer	Honda et al. [26]	Si	FC-72	r	2.2	Not reported	–	Increases 1.8–2.3 times	No correlation
	Chu et al. [7] ^{***}	Si	Water	r	3.6–13.3	Super-hydrophilic (micro/nanostructure)	0°	Max. 2500 kW/m ²	$K = \left(\frac{1+\cos\theta}{16}\right) \left[\frac{2(1+r\cos\theta_{\text{rec}})}{\pi(1+\cos\theta)} + \frac{\pi}{4}(1+\cos\theta)\cos\phi\right]^{1/2}$
	O'Hanley et al. [8]	Si	Water	$R_d/R_q/R_z/r$	0.4–7.2 μm (R_a), 0–1.44 (r)	Hydrophilic and hydrophobic	0–140°	Roughness has no effect	No correlation
	Kim et al. [27]	Si	Water	r	6.46–10.75	Super-hydrophilic (nanostructure)	6.97–1145°	2143 kW/m ²	$q''_{\text{CHF}} = q''_{\text{CHF},\text{Kandlikar}} + Bh_{\text{fg}}\rho_l(1-\alpha)W^2/\lambda_{\text{RT}}^2$
	Guan et al. [18] ^{**}	Si	Pentane, hexane, FC-72	d (hoodoo dia.)	10–80 μm	Highly wetting fluids	–	Increases 1.05–1.67 times	$\frac{q''_{\text{CHF}}}{q''_{\text{CHF},\text{smooth}}} = \sqrt{\frac{A^2 + \sqrt{A^4 + 4}}{2}}$

^{*} θ_r : receding contact angle.
^{**} A: a dimensionless variable including the fluid properties and the wicking velocity.
^{***} K: CHF parameter coming from the relation $q''_{\text{CHF}} = K \times h_{\text{fg}}\rho_v^{1/2} [\sigma g(\rho_l - \rho_v)]^{1/4}$.

enhancement of CHF on silicon wafer surfaces is more pronounced and shows more than a 100% increase in CHF, depending on the roughness factor, r . The considerably higher enhancement of CHF on the silicon surfaces can be attributed to the increased capillary wicking phenomenon with increasing surface roughness. Moreover, because most silicon surfaces were super-hydrophilic, aided by the nanostructures, it is believed that capillary wicking is more intensive in the microgeometries on the silicon surfaces. Because capillary pressure exists if the surface is hydrophilic, the copper surfaces having moderate wettability in the present study certainly have the capillary wicking effect.

It is already discussed that the static and dynamic contact angles viewed in the normal-to-the-scratch direction consistently decrease with increasing R_a because of the capillary wicking (Figs. 5 and 7). Fig. 11 shows the model of a hovering bubble on the modified surface structures created by the current surface roughness following the original model proposed by Haramura and Katto [29]. They explained that CHF occurs at the dry-out when the liquid sublayer beneath the hovering bubble completely evaporates. In the case of a perfectly smooth surface, the liquid amount in the liquid sublayer is established at the initial stage of the bubble cycle without any additional supply. However, for roughened surfaces, adjacent liquid is likely to be transported to the sublayer by capillary wicking. This additional liquid supply could delay the dry-out of the liquid sublayer at the same q'' where CHF occurs at the perfectly smooth surface, thus augmenting the CHF at the roughened surface. Because the capillary wicking increases further with surface roughness, the augmentation of CHF is predicted to increase as well. The CHF measurements in the present study clearly demonstrate this fact with CHF increasing from 775 kW/m² at the smoothest surface ($R_a = 0.041 \mu\text{m}$) to 1625 kW/m² at the roughest surface ($R_a = 2.36 \mu\text{m}$).

The CHF model, including the wettability effect on the conventional hydrodynamic model, was first proposed by Kandlikar [24]. He successfully added the wettability effect in his CHF model by balancing momentum change due to evaporation with the gravity

and surface tension forces exerted on the bubble. Recently, some researchers considered the contribution of surface roughness to CHF as an additional term to Kandlikar's CHF model. Ahn et al. [19] ascribed the large increase of CHF on their super-hydrophilic surfaces to the existence of additional liquid spreading above the wettability effect. This liquid spreading is thought to be another expression of capillary wicking. As shown in Table 2, Chu et al. [7] modified the Kandlikar correlation to accommodate the effect of surface roughness by changing $\cos \theta$ to $r \cos \theta$, where r is the roughness factor. Kim et al. [27] also proposed a CHF correlation including the surface roughness effect by adding a capillary wicking term to the Kandlikar model. Although various attempts were made to fit the capillary effect into the CHF model for explaining the large augmentation on silicon surfaces, an appropriate CHF correlation accounting for the roughness effect on metal surfaces is difficult to analyze. Because a different roughness parameter r is normally used for silicon surfaces, a new correlation using R_a is required to account for the roughness effect on CHF for metal surfaces.

For CHF modeling on the present metal surfaces, it can be thought of as having many small capillary tubes beneath the vapor bubble as shown in Fig. 12. Force balance between momentum change due to evaporation and forces acting on the vapor bubble can then be analyzed in a similar way to Kandlikar's original approach, with an additional capillary force term. Quan et al. [30] first considered capillary wicking due to surface roughness as an additional force term in the force balance equation:

$$F_M = F_{S,1} + F_{S,2} + F_G + F_C \tag{11}$$

where F_M is the momentum change due to the evaporation at the interface and $F_{S,1}$, $F_{S,2}$, F_G , and F_C refer to surface force at the bottom of the bubble, surface force at the top of the bubble, gravity force, and capillary wicking force, respectively.

$$F_M = \frac{1}{\rho_v} \left(\frac{q''_l}{h_{\text{fg}}} \right)^2 H_b \tag{12}$$